Input system:
Dependent variables:x
Independent variables:t
Equations:

$$\frac{dx}{dt} = bx^2 \mathrm{inv}(N + x)$$

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Result:
Dependent variables:s_1, x
Independent variables:t
Additional variables:

$$s_1 = \mathrm{inv}(N + x)$$

Equations:

$$\frac{dx}{dt} = bx^2 s_1$$

$$\frac{ds_1}{dt} = (-1)s_1^3 bx^2$$